	Nickel-Co-Cr alloy, CMSX-4, as cast	Tungsten, commercial purity, R07004, annealed	Molybdenum, 360 grade, fully recrystallized	Tantalum, commercial purity, R05200, cold worked, >99.7% Ta	Niobium, commercial purity, Type 1 (reactor grade)	Molybdenum, Alloy 363, TZM	Niobium-Hf-Ti alloy, Alloy C-103, recrystallized	Tantalum-tungsten alloy, Ta-10W	Nickel-chromium alloy, INCONEL 718, solution treated
General information									
Condition	As cast	Annealed	Fully recrystallized	Cold worked			Recrystallized		Solution treated
UNS number	N13010	R07004	R03600	R05200	R04200	R03630	R04295	R05255	N07718
US name									ASTM Grade N07718; AMS 5662-5664, 5832, 5914, 5962, 5596, 5597, 5950; ASME
EN name									NiCr19NbMo
EN number									2.4668
ISO name									ISO 9723-9725
JIS (Japanese) name									NCF718 to JIS G4901, JIS G4902
Included in Materials Data for Simulation		✓				✓		✓	✓
Materials Data for Simulation name		Tungsten, pure, R07004				Molybdenum alloy, Alloy 363		Tantalum alloy, Ta-10W	Nickel alloy, Inconel 718, solution treated
Composition overview									
Material state	Solid	Solid	Solid	Solid	Solid	Solid	Solid	Solid	Solid
Material class	Metal	Metal	Metal	Metal	Metal	Metal	Metal	Metal	Metal
Material family	Metal (non-ferrous)	Metal (non-ferrous)	Metal (non-ferrous)	Metal (non-ferrous)	Metal (non-ferrous)	Metal (non-ferrous)	Metal (non-ferrous)	Metal (non-ferrous)	Metal (non-ferrous)
Material sub-class	Nickel alloy	Refractory alloy	Refractory alloy	Refractory alloy	Refractory alloy	Refractory alloy	Refractory alloy	Refractory alloy	Nickel alloy
Base material	Ni (Nickel)	W (Tungsten)	Mo (Molybdenum)	Ta (Tantalum)	Nb (Niobium)	Mo (Molybdenum)	Nb (Niobium)	Ta (Tantalum)	Ni (Nickel)
Composition detail (metals, ceramics and glasses)									
Al (aluminum) (%)	5.6	0	0	0	0	0	0	0	0.2 - 0.8
B (boron) (%)	0	0	0	0	0	0	0	0	0 - 0.006
C (carbon) (%)	0	0	0 - 0.03	0	0	0.01 - 0.03	0	0	0 - 0.08
Co (cobalt) (%)	9 - 9.6	0	0	0	0	0	0	0	0 - 1
Cr (chromium) (%)	6.5	0	0	0	0	0	0	0	17 - 21
Cu (copper) (%)	0	0	0	0	0	0	0	0	0 - 0.3
Fe (iron) (%)	0	0	0 - 0.01	0	0	0 - 0.01	0	0	11.1 - 24.6
Hf (hafnium) (%)	0.1	0	0	0	0	0	9 - 11	0	0
Mn (manganese) (%)	0	0	0	0	0	0	0	0	0 - 0.35
Mo (molybdenum) (%)	0.6	0	99.9	0 - 0.02	0	99.3 - 99.5	0	0 - 0.02	2.8 - 3.3
N (nitrogen) (%)	0	0	0 - 0.002	0	0	0 - 0.002	0	0	0
Nb (niobium) (%)	0	0	0	0 - 0.1	99.8	0	85.9 - 90.3	0 - 0.1	2.38 - 2.75
Ni (nickel) (%)	60.7 - 61.7	0	0 - 0.002	0	0	0 - 0.002	0	0	50 - 55
O (oxygen) (%)	0	0	0 - 0.0015	0 - 0.015	0 - 0.015	0 - 0.003	0	0 - 0.015	0
P (phosphorus) (%)	0	0	0	0	0	0	0	0	0 - 0.015
Re (rhenium) (%)	3	0	0	0	0	0	0	0	0
S (sulfur) (%)	0	0	0	0	0	0	0	0	0 - 0.015
Si (silicon) (%)	0	0	0 - 0.01	0	0	0 - 0.01	0	0	0 - 0.35
Ta (tantalum) (%)	6.5	0	0	99.7	0 - 0.1	0	0 - 0.5	88.9 - 91	2.38 - 2.75
Ti (titanium) (%)	1	0	0	0	0	0.4 - 0.55	0.7 - 1.3	0	0.65 - 1.15
W (tungsten) (%)	6 - 6.4	99.9	0	0 - 0.05	0 - 0.03	0	0 - 0.5	9 - 11	0
Zr (zirconium) (%)	0	0	0	0	0	0.06 - 0.12	0 - 0.7	0	0
Other (%)	0	0	0	0 - 0.01	0 - 0.1	0	0 - 0.1	0 - 0.01	0

Price									
Price (USD/kg)	131 - 143	40.2 - 69.2	48.5	375 - 442	127 - 145	50.8	573 - 589	342 - 405	19 - 20.9
Price per unit volume (USD/m^3)	1.14e6 - 1.24e6	774000 - 1.34e6	490000 - 500000	6.19e6 - 7.38e6	1.09e6 - 1.26e6	511000 - 521000	5.04e6 - 5.27e6	5.71e6 - 6.84e6	155000 - 173000
Physical properties									
Density (kg/m^3)	8700	19300 - 19400	10100 - 10300	16500 - 16700	8600 - 8700	10100 - 10300	8800 - 8950	16700 - 16900	8180 - 8260
Mechanical properties									
Young's modulus (Pa)	2.26e11	3.4e11 - 3.5e11	3.15e11 - 3.35e11	1.75e11 - 1.9e11	1e11 - 1.1e11	3.05e11 - 3.25e11	8.2e10 - 9.2e10	1.95e11 - 2.15e11	2.03e11 - 2.13e11
Young's modulus with temperature (Pa) #	2.22e11	3.37e11							1.98e11
Specific stiffness (J/kg)	2.6e7	1.76e7 - 1.81e7	3.08e7 - 3.29e7	1.05e7 - 1.15e7	1.16e7 - 1.27e7	3e7 - 3.21e7	9.23e6 - 1.04e7	1.16e7 - 1.28e7	2.47e7 - 2.6e7
Yield strength (elastic limit) (Pa)	9e8 - 1.15e9	1.35e9 - 1.68e9	1.7e8 - 3.5e8	3.45e8 - 1.06e9	7.5e7 - 9.5e7	7.7e8 - 9.5e8	2.6e8 - 3.8e8	3.8e8 - 5.6e8	7.24e8 - 8e8
Yield strength with temperature (Pa) #	1.02e9	1.4e9							7.7e8
Tensile strength (Pa)	1.21e9	1.67e9 - 2.05e9	3.8e8 - 4.85e8	5.15e8 - 1.4e9	1.7e8 - 1.95e8	8.65e8 - 1.07e9	3.6e8 - 4.5e8	4.8e8 - 6.5e8	8.27e8 - 9.14e8
Tensile strength with temperature (Pa) #	1.21e9								
Specific strength (J/kg)	103000 - 132000	69900 - 87000	16700 - 34300	20800 - 63900	8670 - 11000	75800 - 93600	29300 - 42800	22600 - 33300	88000 - 97400
Elongation (strain)	0.05 - 0.11	0.1 - 0.25	0.2 - 0.45	0.01 - 0.06	0.25 - 0.4	0.08 - 0.12	0.2 - 0.31	0.05 - 0.25	0.35 - 0.5
Tangent modulus (Pa)	3.04e9	3.39e9	9.28e8	5.05e9	4.72e8	1.88e9	7.29e8	9.94e8	1.13e9
Compressive strength (Pa)		1.35e9 - 1.68e9	1.7e8 - 3.5e8	3.45e8 - 1.06e9	7.5e7 - 9.5e7	7.7e8 - 9.5e8	2.6e8 - 3.8e8	3.8e8 - 5.6e8	7.24e8 - 8e8
Flexural modulus (Pa)		3.4e11 - 3.5e11	3.15e11 - 3.35e11	1.75e11 - 1.9e11	1e11 - 1.1e11	3.05e11 - 3.25e11	8.2e10 - 9.2e10	1.95e11 - 2.15e11	2.03e11 - 2.13e11
Flexural strength (modulus of rupture) (Pa)		1.35e9 - 1.68e9	1.7e8 - 3.5e8	3.45e8 - 1.06e9	7.5e7 - 9.5e7	7.7e8 - 9.5e8	2.6e8 - 3.8e8	3.8e8 - 5.6e8	7.24e8 - 8e8
Shear modulus (Pa)		1.3e11 - 1.38e11	1.2e11 - 1.3e11	6.4e10 - 7.2e10	3.4e10 - 4e10	1.15e11 - 1.3e11	2.5e10 - 3.5e10	7e10 - 8.5e10	7.9e10 - 8.31e10
Shear modulus with temperature (Pa) #									8.12e10
Bulk modulus (Pa)		2.46e11 - 2.78e11	2.5e11 - 2.74e11	1.82e11 - 2.28e11	1.5e11 - 1.84e11	2.4e11 - 2.65e11	1.05e11 - 1.4e11	1.9e11 - 2.6e11	1.61e11 - 1.69e11
Poisson's ratio	0.389	0.27 - 0.29	0.29 - 0.295	0.34 - 0.36	0.39 - 0.4	0.29 - 0.295	0.37 - 0.39	0.33 - 0.36	0.29 - 0.302
Shape factor	18	18	30	22	30	24	21	26	21
Hardness - Vickers (HV)		450 - 530	150 - 200	130 - 310	80 - 110	260 - 360	160 - 240	160 - 240	400 - 500
Hardness - Rockwell B (HRB)									112 - 117
Hardness - Rockwell C (HRC)									41 - 49
Hardness - Brinell (HB)									379 - 471
Elastic stored energy (springs) (Pa)	1.82e6 - 2.89e6	2.67e6 - 4.05e6	49000 - 171000	400000 - 2.51e6	27100 - 42700	949000 - 1.42e6	400000 - 809000	363000 - 744000	1.26e6 - 1.54e6
Fatigue strength at 10^7 cycles (Pa)		6.65e8 - 1.02e9	1.5e8 - 2.45e8	2.05e8 - 2.5e8	6.5e7 - 1e8	3.45e8 - 5.35e8	1.4e8 - 2.25e8	3e8 - 4e8	4.85e8 - 7.55e8
Fatigue strength model (stress amplitude) (Pa) #		6.24e8 - 1.09e9	1.39e8 - 2.65e8	9.84e7 - 5.21e8	6.3e7 - 1.03e8	3.23e8 - 5.71e8	1.31e8 - 2.41e8	2.54e8 - 4.72e8	4.77e8 - 7.67e8
Impact & fracture properties									
Fracture toughness (Pa.m^0.5)		1.2e8 - 1.5e8	2e7 - 4e7	9e7 - 1.5e8	9e7 - 1.2e8	3e7 - 5e7	1.1e8 - 1.5e8	1.2e8 - 1.5e8	2.44e8 - 2.71e8
Toughness (G) (N/m)		42200 - 64500	1350 - 4500	46800 - 117000	78400 - 135000	3010 - 7540	142000 - 255000	70800 - 109000	285000 - 352000
Thermal properties									
Melting point (°C)		3410 - 3420	2610 - 2620	2990 - 3000	2460 - 2470	2610 - 2620	2340 - 2360	3030 - 3040	1260 - 1340
Maximum service temperature (°C)		825 - 1210	867 - 1310	1370 - 1980	547 - 737	867 - 1310	520 - 700	1370 - 2480	632 - 705
Minimum service temperature (°C)		-273	-273	-273	-273	-273	-273	-273	-273
Thermal conductivity (W/m.°C)	7.7	170 - 175	129 - 147	54 - 58	54 - 57	129 - 147	40 - 44	34 - 52	11.6 - 12.6
Thermal conductivity with temperature (W/m.°C) #		173							11.3
Specific heat capacity (J/kg.°C)	219	135 - 141	255 - 275	138 - 144	260 - 275	255 - 275	330 - 350	135 - 145	440 - 458
Specific heat capacity with temperature (J/kg.°C) #		129							423
Thermal expansion coefficient (strain/°C)	0.0000113	0.0000042 - 0.0000044	0.0000048 - 0.0000055	0.0000064 - 0.0000066	0.000007 - 0.0000072	0.0000047 - 0.0000055	0.0000078 - 0.0000084	0.000005 - 0.0000065	0.0000128 - 0.0000134
Thermal expansion coefficient with temperature (strain/°C) #		0.00000443						0.0000061	0.0000123
Thermal expansion coefficient with temperature_Reference temp (°C)		20						20	20

Thermal shock resistance (°C)	352 - 450	908 - 1140	101 - 211	291 - 895	99.8 - 129	468 - 610	366 - 546	332 - 510	263 - 297
Thermal distortion resistance (W/m)	681000	3.9e7 - 4.12e7	2.44e7 - 2.94e7	8.28e6 - 8.95e6	7.58e6 - 8.06e6	2.45e7 - 3e7	4.88e6 - 5.5e6	6.12e6 - 9.64e6	879000 - 966000
Latent heat of fusion (J/kg)		190000 - 195000	285000 - 292000	145000 - 174000	290000 - 320000	280000 - 300000	270000 - 300000	145000 - 180000	275000 - 300000
Electrical properties									
Electrical resistivity (ohm.m)	0.00000133	5.4e-8 - 6e-8	5.2e-8 - 6e-8	1.13e-7 - 1.36e-7	1.3e-7 - 1.6e-7	5.2e-8 - 6e-8	1.5e-7 - 2e-7	1.6e-7 - 2e-7	0.00000115 - 0.00000125
Electrical resistivity with temperature (ohm.m) #		5.42e-8							
Electrical conductivity (Siemens/m)	754000	1.66e7 - 1.85e7	1.66e7 - 1.93e7	7.37e6 - 8.87e6	6.26e6 - 7.71e6	1.66e7 - 1.93e7	5e6 - 6.67e6	5e6 - 6.26e6	800000 - 870000
Electrical conductivity with temperature (Siemens/m) #		1.85e7							
Temperature dependence of resistivity (/°C)		20.4 - 20.6			10 - 10.4				
Galvanic potential (V)		-0.39 - -0.31	-0.2 - -0.12	0 - 0.08	-0.3 - -0.22	-0.2 - -0.12	-0.27 - -0.19	-0.04 - 0.04	-0.05 - 0.03
Magnetic properties									
Magnetic type	Non-magnetic	Non-magnetic	Non-magnetic	Non-magnetic	Non-magnetic	Non-magnetic	Non-magnetic	Non-magnetic	Non-magnetic
Optical, aesthetic and acoustic properties									
Transparency		Opaque	Opaque	Opaque	Opaque	Opaque	Opaque	Opaque	Opaque
Acoustic velocity (m/s)	5100	4200 - 4260	5550 - 5740	3250 - 3380	3400 - 3570	5480 - 5660	3040 - 3220	3410 - 3580	4970 - 5100
Mechanical loss coefficient (tan delta)		0.00006 - 0.0001	0.0004 - 0.001	0.0001 - 0.0005	0.002 - 0.0027	0.0001 - 0.0003	0.0005 - 0.0009	0.0002 - 0.0004	0.0001 - 0.0003
Restricted substances risk indicators									
SIN List indicator (0-1, 1 = high risk)	0.1	0	0	0	0	0	0	0	0.01
Critical materials risk									
Contains >5wt% critical elements?	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Processing properties									
Metal casting		Unsuitable	Unsuitable	Unsuitable	Unsuitable	Unsuitable	Unsuitable	Unsuitable	Unsuitable
Metal cold forming		Limited use	Acceptable	Excellent	Excellent	Acceptable	Excellent	Excellent	Acceptable
Metal hot forming		Limited use	Acceptable	Excellent	Excellent	Acceptable	Excellent	Excellent	Limited use
Metal press forming		Limited use	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable
Metal deep drawing		Limited use	Limited use	Limited use	Limited use	Limited use	Limited use	Limited use	Limited use
Machining speed (m/s)		0.0559	0.325	0.132	0.66	0.178	0.3	0.152	0.0559
Weldability		Good	Good	Good	Good	Excellent	Good	Poor	Good
Weldability_Notes		Preheating and post weld heat treatments are required	Preheating is not required, post weld heat treatment is required	Preheating and post weld heat treatments are required	Preheating and post weld heat treatments are required	Preheating is not required, post weld heat treatment is required	Preheating and post weld heat treatments are required	Preheating and post weld heat treatments are required	Preheating is not required, post weld heat treatment is required
Weldability - MIG									Good
Weldability - plasma									Good
Weldability - SAW									Good
Weldability - TIG									Good
Brazeability									Good
Durability									
Water (fresh)		Excellent	Excellent	Excellent	Excellent	Excellent	Excellent	Excellent	Excellent
Water (salt)		Acceptable	Acceptable	Excellent	Excellent	Acceptable	Excellent	Excellent	Excellent
Weak acids		Excellent	Acceptable	Excellent	Excellent	Acceptable	Excellent	Excellent	Excellent
Strong acids		Limited use	Unacceptable	Acceptable	Acceptable	Unacceptable	Acceptable	Acceptable	Excellent
Weak alkalis		Acceptable	Acceptable	Excellent	Excellent	Acceptable	Excellent	Excellent	Excellent
Strong alkalis		Limited use	Limited use	Excellent	Excellent	Limited use	Limited use	Excellent	Excellent
Organic solvents		Excellent	Excellent	Excellent	Excellent	Excellent	Excellent	Excellent	Excellent
Oxidation at 500C		Excellent	Acceptable	Acceptable	Unacceptable	Acceptable	Limited use	Acceptable	Excellent

UV radiation (sunlight)		Excellent	Excellent	Excellent	Excellent	Excellent	Excellent	Excellent	Excellent
Galling resistance (adhesive wear)		Limited use	Limited use	Unacceptable	Unacceptable	Limited use	Unacceptable	Unacceptable	Limited use
Flammability		Non-flammable	Non-flammable	Non-flammable	Non-flammable	Non-flammable	Non-flammable	Non-flammable	Non-flammable
Corrosion resistance of metals									
Stress corrosion cracking		Slightly susceptible	Not susceptible	Susceptible	Not susceptible	Not susceptible	Not susceptible	Slightly susceptible	Not susceptible
Stress corrosion cracking_Notes		Rated in chloride; May be susceptible in halide, ammonia, nitrogen, acidic, caustic, carbonate environments	Rated in chloride; May be susceptible in halide, ammonia, nitrogen, acidic, caustic, carbonate environments	Rated in chloride; Other susceptible environments: Hydrogen sulfide	Rated in chloride; Other susceptible environments: Hydroxide	Rated in chloride; May be susceptible in halide, ammonia, nitrogen, acidic, caustic, carbonate environments	Rated in chloride; Other susceptible environments: Hydroxide	Rated in chloride; Other susceptible environments: Moist air, water, chloride, sulfate, sulfide	Rated in chloride; Other susceptible environments: Hydroxide
Intergranular (weld line) corrosion resistance									Excellent
Inorganic acids									Excellent
Organic acids									Excellent
Alkalis									Excellent
Humidity / water									Excellent
Sea water									Excellent
Sour oil and gas									Excellent
Environmental indicators - material production									
Climate change (CO2-eq) (virgin grade production) (kg/kg)	56.2 - 61.9	36.2 - 39.9	16.2 - 17.9	166 - 184	117 - 129	16.4 - 18	111 - 123	153 - 169	15.7 - 17.3
Climate change (CO2-eq) (typical grade production) (kg/kg)	34.2 - 37.7	29.4 - 32.4	14.2 - 15.7	153 - 170	107 - 118	14.4 - 15.8	103 - 114	139 - 154	10.5 - 11.6
Embodied energy (virgin grade production) (J/kg)	8.06e8 - 8.89e8	5.5e8 - 6.06e8	1.49e8 - 1.65e8	2.56e9 - 2.82e9	1.76e9 - 1.94e9	1.52e8 - 1.68e8	1.68e9 - 1.85e9	2.36e9 - 2.6e9	2.21e8 - 2.43e8
Embodied energy (typical grade production) (J/kg)	4.69e8 - 5.17e8	4.29e8 - 4.72e8	1.28e8 - 1.41e8	2.33e9 - 2.56e9	1.59e9 - 1.75e9	1.3e8 - 1.44e8	1.53e9 - 1.69e9	2.11e9 - 2.33e9	1.43e8 - 1.57e8
Water usage (m^3/kg)		0.147 - 0.162	0.341 - 0.377	0.616 - 0.681	0.147 - 0.162	0.34 - 0.376	0.917 - 1.01	0.569 - 0.629	0.235 - 0.259

Environmental indicators - processing								
Metal casting, climate change (CO2-eq) (kg/kg)	2.99 - 3.35		2.82 - 3.22	3.46 - 3.96	3.48 - 3.98	3.61 - 4.14	2.82 - 3.25	
Metal casting, embodied energy (J/kg)	4.23e7 - 4.73e7		4.03e7 - 4.56e7	4.81e7 - 5.47e7	4.83e7 - 5.49e7	4.98e7 - 5.69e7	4.02e7 - 4.6e7	
Metal rolling and forging, climate change (CO2-eq) (kg/kg)	0.955 - 1.29	0.273 - 0.544	0.325 - 0.961	0.169 - 0.218	1.04 - 1.37	0.435 - 0.664	0.348 - 0.532	1.19 - 1.43
Metal rolling and forging, embodied energy (J/kg)	1.46e7 - 1.97e7	4.17e6 - 8.3e6	4.97e6 - 1.47e7	2.58e6 - 3.32e6	1.59e7 - 2.1e7	6.66e6 - 1.01e7	5.33e6 - 8.11e6	1.82e7 - 2.19e7
Metal extrusion, hot, climate change (CO2-eq) (kg/kg)	1.84 - 2.5	0.485 - 1.02	0.588 - 1.84	0.279 - 0.37	2 - 2.66	0.808 - 1.25	0.635 - 0.993	2.31 - 2.78
Metal extrusion, hot, embodied energy (J/kg)	2.81e7 - 3.81e7	7.41e6 - 1.55e7	9e6 - 2.81e7	4.26e6 - 5.64e6	3.06e7 - 4.07e7	1.24e7 - 1.91e7	9.71e6 - 1.52e7	3.52e7 - 4.24e7
Wire drawing, climate change (CO2-eq) (kg/kg)	0.859 - 0.95	0.859 - 0.95	0.859 - 0.95	0.859 - 0.95	0.859 - 0.95	0.859 - 0.95	0.859 - 0.95	0.859 - 0.95
Wire drawing, embodied energy (J/kg)	1.31e7 - 1.45e7	1.31e7 - 1.45e7	1.31e7 - 1.45e7	1.31e7 - 1.45e7	1.31e7 - 1.45e7	1.31e7 - 1.45e7	1.31e7 - 1.45e7	1.31e7 - 1.45e7
Metal powder forming, climate change (CO2-eq) (kg/kg)	3.85 - 4.44	5.46 - 6.52	3.47 - 3.99	5.25 - 6.15	5.47 - 6.52	6.3 - 7.43	3.44 - 4.07	4.55 - 5.53
Metal powder forming, embodied energy (J/kg)	4.73e7 - 5.46e7	6.69e7 - 8e7	4.27e7 - 4.92e7	6.44e7 - 7.54e7	6.7e7 - 7.99e7	7.71e7 - 9.1e7	4.23e7 - 5.02e7	5.59e7 - 6.79e7
Chemical vapor deposition (CVD), climate change (CO2-eq) (kg/kg)	579 - 657	869 - 984	442 - 586	884 - 1080	854 - 1010	823 - 1010	442 - 606	839 - 1010
Chemical vapor deposition (CVD), embodied energy (J/kg)	7.04e9 - 7.99e9	1.06e10 - 1.2e10	5.37e9 - 7.13e9	1.07e10 - 1.31e10	1.04e10 - 1.23e10	1e10 - 1.23e10	5.37e9 - 7.37e9	1.02e10 - 1.23e10
Machining, coarse, climate change (CO2-eq) (kg/kg)	1.22 - 1.42	1.02 - 1.2	1.03 - 1.32	0.988 - 1.1	1.25 - 1.45	1.07 - 1.23	1.04 - 1.19	1.29 - 1.46
Machining, coarse, embodied energy (J/kg)	1.87e7 - 2.17e7	1.56e7 - 1.83e7	1.58e7 - 2.02e7	1.51e7 - 1.68e7	1.91e7 - 2.21e7	1.63e7 - 1.88e7	1.59e7 - 1.82e7	1.98e7 - 2.23e7
Machining, fine, climate change (CO2-eq) (kg/kg)	1.76 - 2.15	1.15 - 1.49	1.19 - 1.86	1.05 - 1.19	1.83 - 2.23	1.29 - 1.59	1.22 - 1.47	1.97 - 2.28
Machining, fine, embodied energy (J/kg)	2.69e7 - 3.29e7	1.75e7 - 2.27e7	1.83e7 - 2.84e7	1.61e7 - 1.82e7	2.8e7 - 3.4e7	1.98e7 - 2.43e7	1.86e7 - 2.25e7	3.01e7 - 3.48e7
Machining, grinding, climate change (CO2-eq) (kg/kg)	2.29 - 2.89	1.28 - 1.77	1.35 - 2.39	1.12 - 1.29	2.42 - 3.01	1.52 - 1.95	1.39 - 1.75	2.65 - 3.1
Machining, grinding, embodied energy (J/kg)	3.51e7 - 4.41e7	1.95e7 - 2.7e7	2.07e7 - 3.65e7	1.71e7 - 1.96e7	3.7e7 - 4.59e7	2.32e7 - 2.98e7	2.12e7 - 2.68e7	4.04e7 - 4.73e7

Machining, laser, CO2, climate change (CO2-eq) (kg/m)		0.069 - 0.0782	0.103 - 0.117	0.0526 - 0.0698	0.105 - 0.128	0.102 - 0.12	0.098 - 0.12	0.0526 - 0.0722	0.0998 - 0.12
Machining, laser, CO2, embodied energy (N)		838000 - 951000	1.26e6 - 1.42e6	639000 - 848000	1.28e6 - 1.56e6	1.23e6 - 1.46e6	1.19e6 - 1.46e6	639000 - 877000	1.21e6 - 1.46e6
Recycling and end of life									
Recycle	✓	✓	✓	✓	✓	✓	✓	✓	✓
Functional recycle	✗	✓	✓	✓	✓	✓	✗	✗	✗
Climate change (CO2-eq), recycling (kg/kg)	7.31 - 8.05	21.4 - 23.5	9.56 - 10.6	97.9 - 109	69 - 76.1	9.68 - 10.6	65.5 - 72.6	90.3 - 99.7	2.04 - 2.25
Embodied energy, recycling (J/kg)	5.64e7 - 6.22e7	2.86e8 - 3.15e8	7.75e7 - 8.58e7	1.33e9 - 1.47e9	9.15e8 - 1.01e9	7.9e7 - 8.74e7	8.74e8 - 9.62e8	1.23e9 - 1.35e9	1.55e7 - 1.7e7
Recycle fraction in current supply (%)	45	46	30	19	20	30	18	22	38
Combust for energy recovery	✗	✗	✗	✗	✗	✗	✗	✗	✗
Landfill	✗	✓	✓	✓	✓	✓	✓	✓	✗
Biodegrade	✗	✗	✗	✗	✗	✗	✗	✗	✗
Links									
PMP records	0	0	0	0	0	0	3	0	0
ProcessUniverse	112	92	86	80	86	87	86	84	112
Producers	17	9	4	6	5	4	4	5	22
Reference	33	33	22	25	24	20	24	21	52
Shape	24	24	24	24	24	24	24	24	24
Full Datasheet	view	view	view	view	view	view	view	view	view

#Functional Parameters	
Temperature (°C)	23
Stress Ratio	-1
Number of Cycles (cycles)	1e7